

3 Entropy

In this section we introduce the function of state S , the entropy for reversible processes. We then discuss irreversible processes and a number of examples to illustrate entropy in use.

3.1 Clausius' theorem

The conclusion from the last section was that a reversible cycle (with efficiency η_{rev}) provides an upper limit to the efficiency of a heat engine running between reservoirs at two temperatures, i.e. for a general (not necessarily reversible) such cycle with efficiency η

$$\eta_{\text{rev}} \geq \eta \quad \Rightarrow \quad 1 - \frac{Q_2^{\text{rev}}}{Q_1^{\text{rev}}} \geq 1 - \frac{Q_2}{Q_1} \quad \Rightarrow \quad \frac{Q_2}{Q_1} \geq \frac{Q_2^{\text{rev}}}{Q_1^{\text{rev}}} = \frac{T_2}{T_1} .$$

Thus

$$\frac{Q_2}{T_2} \geq \frac{Q_1}{T_1} \Leftrightarrow \frac{Q_1}{T_1} + \frac{Q_2}{T_2} \leq 0 \quad \Rightarrow \quad \sum_{\text{stages}} \frac{Q}{T} \leq 0 , \quad (3.1)$$

where we have now used a convention that heat **into** the body is **positive**, and the summation is a prelude to the more general case.

We will now construct a similar result in general, where the number of degrees of freedom may be more than two (i.e. more than p and V , for example) and the cycle may be of any complexity.

To do this we will need *three* bodies.

- (i) The body γ , following the arbitrary cycle (e.g. in the p - V plane, but may involve other parameters),
- (ii) A single large thermal reservoir, Γ , at a fixed temperature, T_0 .
- (iii) And finally a third body, γ' which carries heat from the reservoir, Γ , to the body γ . γ' undergoes a Carnot cycle between T_0 (i.e. Γ) and T , the temperature at which it deposits heat to γ .

I think, although not expressed in the literature, that we must assume the following inequalities about the heat capacities of the three bodies: $C_\gamma \ll C_{\gamma'} \ll C_\Gamma$.

So for a particular small (infinitesimal) element of the cycle of γ , the process is:

- (i) γ is in its initial state; γ' is at temperature T_0 ;
- (ii) γ' is brought adiabatically to T ;
- (iii) γ moves to its final state (for this Carnot cycle by γ'), absorbing dQ from γ' , which proceeds along an isotherm at T (this is where one must assume something about the relative magnitudes of the heat capacities for γ and γ');
- (iv) γ' is returned adiabatically to T_0 , compressed or expanded as required to reach its initial state.

Since γ' undergoes a Carnot cycle, and it deposits dQ at temperature T , it must gain from the reservoir, Γ , heat $q = dQT_0/T$ at stage (iv) above (from $q/T_0 = dQ/T$ for a Carnot cycle). **Note** $q > 0$ implies that heat is **donated** to γ from the reservoir.

To allow for γ' to deposit heat at a well-defined temperature, T , to γ , which might not be in equilibrium, requires $C_\gamma \ll C_{\gamma'}$.

Thus over the whole cycle, the reservoir loses heat $\oint q = T_0 \oint dQ/T$ (noting that T_0 is constant) as γ performs its (perhaps complicated) cycle. Since at the end of the cycle γ and γ' are returned to their initial states, their internal energies are the same as their initial values.

The role of γ' is to be both a hot and a cold reservoir for γ , depending on which part of γ 's cycle γ is absorbing or donating energy to γ' . Work may be done on both γ , say W' and on γ , as W .

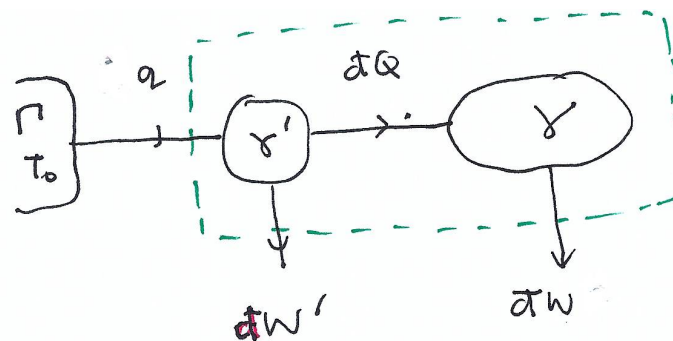


Figure 3.1: The green dashed line indicates the composite engine made up of the Carnot engine, γ' , and the body undergoing a general cycle, γ .

Let us now regard γ' and γ to be a composite "engine", as in Fig. (3.1). It looks a little strange as γ replaces the "low-temperature" reservoir and γ' is run as a heat engine or refrigerator depending on when it is donating heat or receiving heat from γ . The dashed line indicates the extent of the composite engine.

The composite engine is Kelvin-impossible (running over a complete cycle) if $\oint q > 0$ as then Γ is providing heat which turns into net work, $W' + W > 0$. So in that case the Second Law is contradicted. On the other hand, if $\oint q < 0$ then the external agencies are doing work on the combination of γ and γ' , and it is being turned into heat and donated to the reservoir Γ . This is allowed by the Second Law.

Thus we may deduce, since $T_0 > 0$,

$$\oint q = \oint \mathrm{d}Q/T \leq 0, \quad (3.2)$$

which is *Clausius' Inequality*.

But if the cycle of γ is *reversible*, then there is a cycle with $\mathrm{d}Q \rightarrow -\mathrm{d}Q$, providing the opposite sign of Clausius inequality. Thus since both are true, for a reversible cycle

$$\oint \frac{\mathrm{d}Q}{T} = 0. \quad (3.3)$$

We will initially focus on the consequences of Clausius' Equality for reversible processes, Eq. (3.3). It implies the integral of $\mathrm{d}Q/T$ along any two paths, say P and P' , between A and B (e.g. in the p - V plane) must be equal, as if one adds one path to the other one reversed, the sum is a closed path and the integral is zero (Eq. (3.3)):

$$\int_A^B \frac{\mathrm{d}Q}{T} + \int_B^A \frac{\mathrm{d}Q}{T} = \oint \frac{\mathrm{d}Q}{T} = 0 \Rightarrow \int_A^B \frac{\mathrm{d}Q}{T} = - \int_B^A \frac{\mathrm{d}Q}{T} = \int_A^B \frac{\mathrm{d}Q}{T},$$

i.e. path independence.

This implies that the integral between A and B is path-independent and hence only depends on the end points, so we may introduce a function of the endpoints, the *entropy*, S , and now not specifying a path

$$S_B - S_A = \int_A^B \frac{\mathrm{d}Q}{T},$$

or in differential form

$$\mathrm{d}S = \frac{\mathrm{d}Q}{T}, \quad (3.4)$$

with S **being a function of state** as the path does not affect its value. This implies that $1/T$ is an "integrating factor" to turn $\mathrm{d}Q$ into a total differential – as in preliminary II in section 2.

If we now consider a cycle composed to two legs, one irreversible, from A to B , closed by the second leg taking the system back to A by a reversible process, we have from Clausius (here

T_{ext} is the temperature of the body γ' in the Carnot cycle transferring heat to the body γ .)

$$\oint \frac{dQ}{T} \leq 0 \quad (3.5)$$

$$\Rightarrow \int_A^B \frac{dQ}{T_{\text{ext}}} + \int_B^A \frac{dQ^{\text{rev}}}{T} \leq 0$$

$$\Rightarrow S_B - S_A \geq \int_A^B \frac{dQ}{T_{\text{ext}}} \quad (3.6)$$

Notice if a system is contained in adiabatic walls, so that $dQ = 0$, then

$$S_B - S_A \geq 0 \quad \text{or} \quad \Delta S \geq 0, \quad (3.7)$$

i.e. for an isolated system the entropy cannot decrease.

Let us return to the First Law to see the effect of the above discussion of reversible processes. Eq. (3.4) implies that for *reversible* changes $dQ^{\text{rev}} = TdS$. Thus for reversible changes the First Law may be written as

$$dU = dQ^{\text{rev}} - p dV$$

$$\Rightarrow dU = TdS - p dV. \quad (3.8)$$

3.2 Examples of calculating entropy changes

1. Blocks coming into contact

Two identical blocks of metal, each of heat capacity C and temperatures $T_1^{(0)}$ and $T_2^{(0)}$ (with $T_2^{(0)} > T_1^{(0)}$), are brought into contact. We assume that there is no change in volume and no work performed. What is the change of entropy of each block? Of the Universe (everything else)?

First we need to calculate the final temperature, T_f . Assume the whole system is isolated, so that no energy is lost to the environment.

$$0 = \Delta U^{\text{tot}} = C(T_f - T_1^{(0)}) + C(T_f - T_2^{(0)})$$

$$\Rightarrow T_f = \frac{T_1^{(0)} + T_2^{(0)}}{2}. \quad (3.9)$$

Next we calculate the change in entropy. When the blocks come into contact they are no longer in equilibrium, making a direct solution difficult. Instead we imagine a transformation that makes the heat transport sufficiently slow that the blocks remain in equilibrium within themselves as, if their relaxation time is τ ,

$$\frac{\partial T_1}{\partial t} \tau \ll T_1 \quad \text{and} \quad \frac{\partial T_2}{\partial t} \tau \ll T_2,$$

so the fractional change of temperature in τ is very small and thus the blocks should be internally in equilibrium.

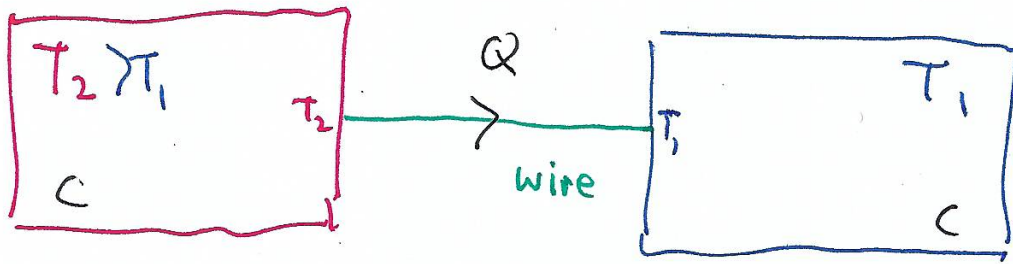


Figure 3.2: A wire of very low thermal conductivity connecting two blocks, allowing them to be at two different temperatures, T_1 and T_2 , but simultaneously in local equilibrium.

One way to achieve this is to couple them by a badly thermally conducting wire, see Fig. (3.2).

Now the transformation of the blocks is reversible, $\delta Q = C dT$ and

$$\Delta S_1 = \int_{T_1}^{T_f} \frac{\delta Q}{T} = \int_{T_1}^{T_f} \frac{C dT}{T} = C \ln \left(\frac{T_f}{T_1} \right) > 0$$

(Similarly:) $\Delta S_2 = C \ln \left(\frac{T_f}{T_2} \right) < 0$.

Since the blocks are isolated, the change of entropy of the Universe is the sum

$$\Delta S_{\text{Univ}} = \Delta S_1 + \Delta S_2 = C \ln \left(\frac{T_f^2}{T_1 T_2} \right)$$

(Use Eq. (3.9).) $= C \ln \left[\frac{(T_1 + T_2)^2}{4T_1 T_2} \right] > 0$.

To see the inequality, note

$$(T_1 + T_2)^2 \geq (T_1 + T_2)^2 - (T_1 - T_2)^2 = 4T_1 T_2.$$

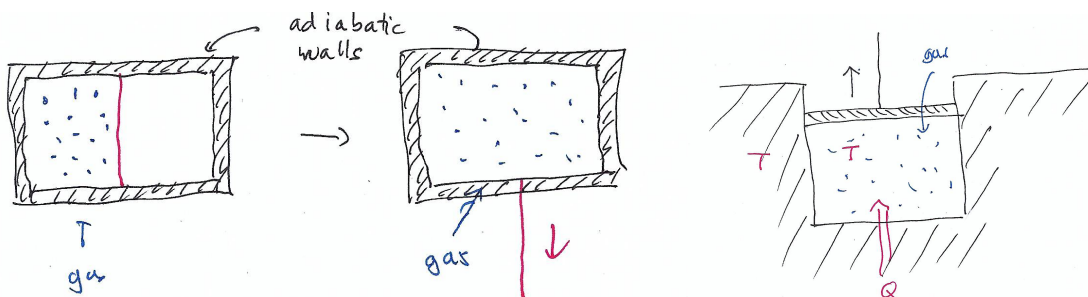


Figure 3.3: Left: the initial configuration with gas confined to the left part of container by a membrane. Centre: the membrane has been removed doing no work. Right: the thought experiment.

2. Joule's second experiment

The membrane in the left hand figure in Fig. (3.3) is slid sideways to no work is

performed on the gas (mid image in Fig. (3.3)), which then fills both parts of the container with adiabatic walls, isolating it from the rest of the Universe. What is the change of entropy?

We may see without calculation that the transformation is irreversible. Since the system is isolated, $\Delta S_{\text{system}} > 0$. Since the system is isolated,

$$\Delta S_{\text{Universe}} = \Delta S_{\text{system}} > 0 .$$

To calculate the increase of the entropy of the gas, assumed to be ideal, we imagine (i.e. thought experiment) a reversible transformation taking the gas between the same initial and final states. We consider an isothermal expansion, with the heat bath providing the energy to ensure the temperature remains constant. See the right hand figure in Fig. (3.3),

$$\begin{aligned} dU &= dW^{\text{rev}} + dQ^{\text{rev}} = 0 \\ \Rightarrow dQ^{\text{rev}} &= -dW^{\text{rev}} \\ &= p dV . \\ \Rightarrow \Delta S_{\text{system}} &= \int \frac{dQ^{\text{rev}}}{T} = nR \int_{V_1}^{V_2} \frac{dV}{V} = nR \ln \left(\frac{V_2}{V_1} \right) > 0 . \end{aligned}$$

Note this transformation is reversible so $\Delta S_{\text{Universe}} = 0$.

ΔS for ...	Real experiment	Thought experiment
System	$nR \ln(V_2/V_1)$	$nR \ln(V_2/V_1)$
Surroundings	0	$-nR \ln(V_2/V_1)$
Universe	$nR \ln(V_2/V_1)$	0

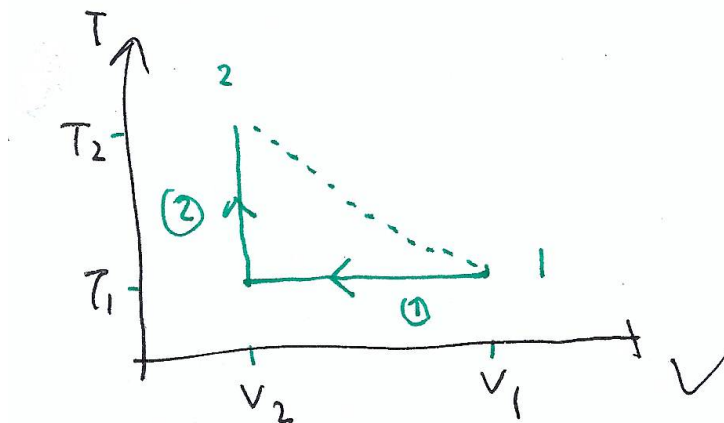


Figure 3.4: The isotherm and isochore, in the T - V plane.

3. Entropy of ideal gas

The last example can be generalised to an arbitrary transformation (for an ideal gas) from (T_1, V_1) to (T_2, V_2) , see Fig. (3.4). This is discussed in more detail in problem (2.4).

We may imagine going from 1 to 2 using an isotherm, (1), followed by an isochore, (2).

The same calculation as before yields

$$\Delta S_{\textcircled{1}} = nR \ln \left(\frac{V_2}{V_1} \right) . \quad (3.10)$$

Along the isochore, $\textcircled{2}$,

$$\begin{aligned} dQ^{\text{rev}} &= C_v dT \\ \Rightarrow \Delta S_{\textcircled{2}} &= \int C_v \frac{dT}{T} = C_v \ln \left(\frac{T_2}{T_1} \right) . \end{aligned} \quad (3.11)$$

Therefore the change in entropy after following both paths is:

$$\Delta S^{\text{tot}} = \Delta S_{\textcircled{1}} + \Delta S_{\textcircled{2}} = nR \ln \left(\frac{V_2}{V_1} \right) + C_v \ln \left(\frac{T_2}{T_1} \right) .$$

If the specific heat for an ideal gas is $C_v = (m/2)nR = N(m/2)k_B$, where m is the number of degrees of freedom of each molecule and N is the total number of molecules, then

$$\frac{\Delta S^{\text{tot}}}{Nk_B} = \ln \left[\left(\frac{V_2}{V_1} \right) \left(\frac{T_2}{T_1} \right)^{m/2} \right] .$$

3.3 Summary

- During spontaneous processes, the entropy of the Universe increases (but it is possible for the entropy of the system to decrease).
- During a reversible transformation, the entropy of the Universe remains constant (but the entropy of the system may not).
- Spontaneous processes correspond to an increase of the entropy of the Universe and therefore are irreversible.
- A transformation which is both reversible and adiabatic is isentropic. The entropy of the system remains the same.